Pre-eclampsia and eclampsia

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Outline

Epidemiology

About 10% of pregnancies involve hypertension, most being mild gestational hypertension.
Pre-eclampsia affects ~3% of pregnancies and is a leading cause of maternal death [50,000–75,000 annually worldwide], and is often linked to fetal growth restriction and high perinatal complications.

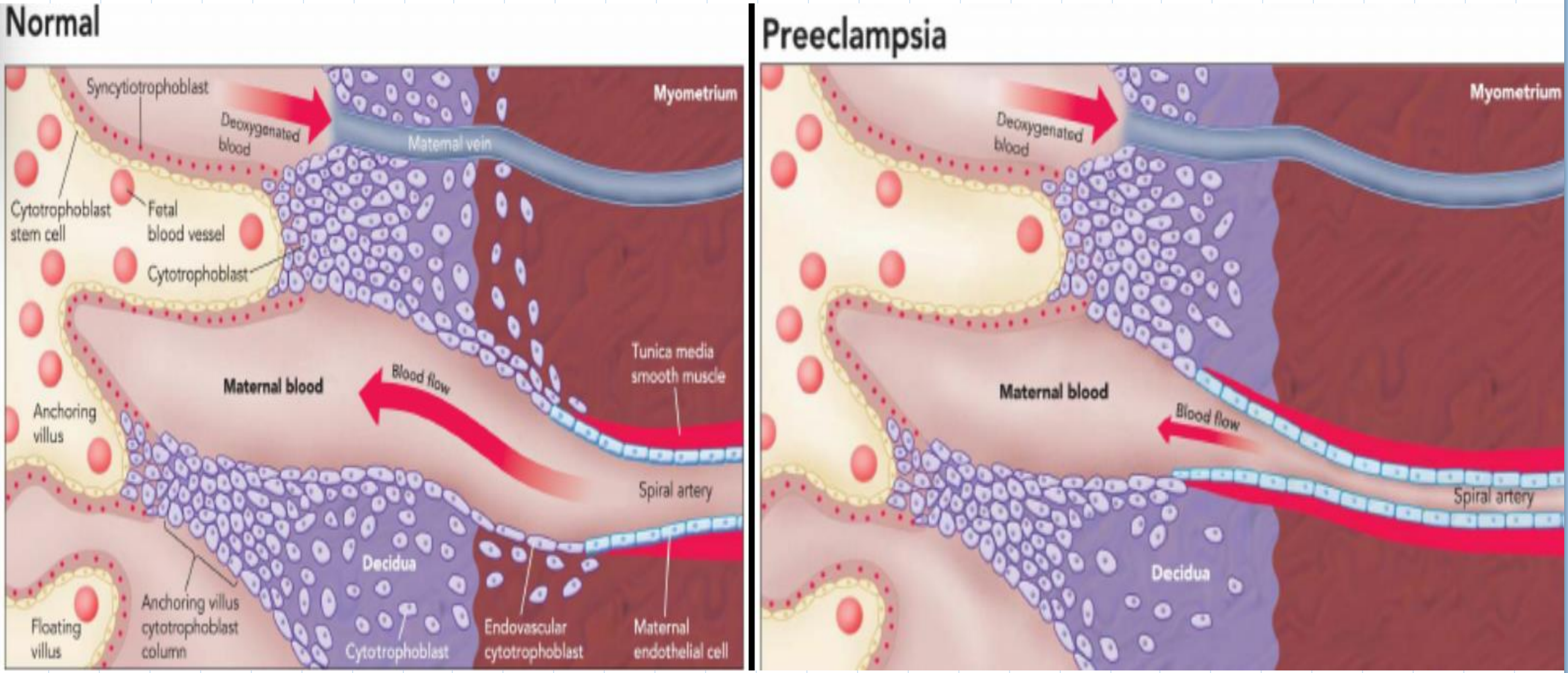
Classification

- Hypertensive disorders in pregnancy (HDP)
  - < 20 weeks of gestation
    - Chronic hypertension
  - ≥20 weeks of gestation
    - Preeclampsia
    - Gestational hypertension
    - Chronic hypertension with superimposed preeclampsia

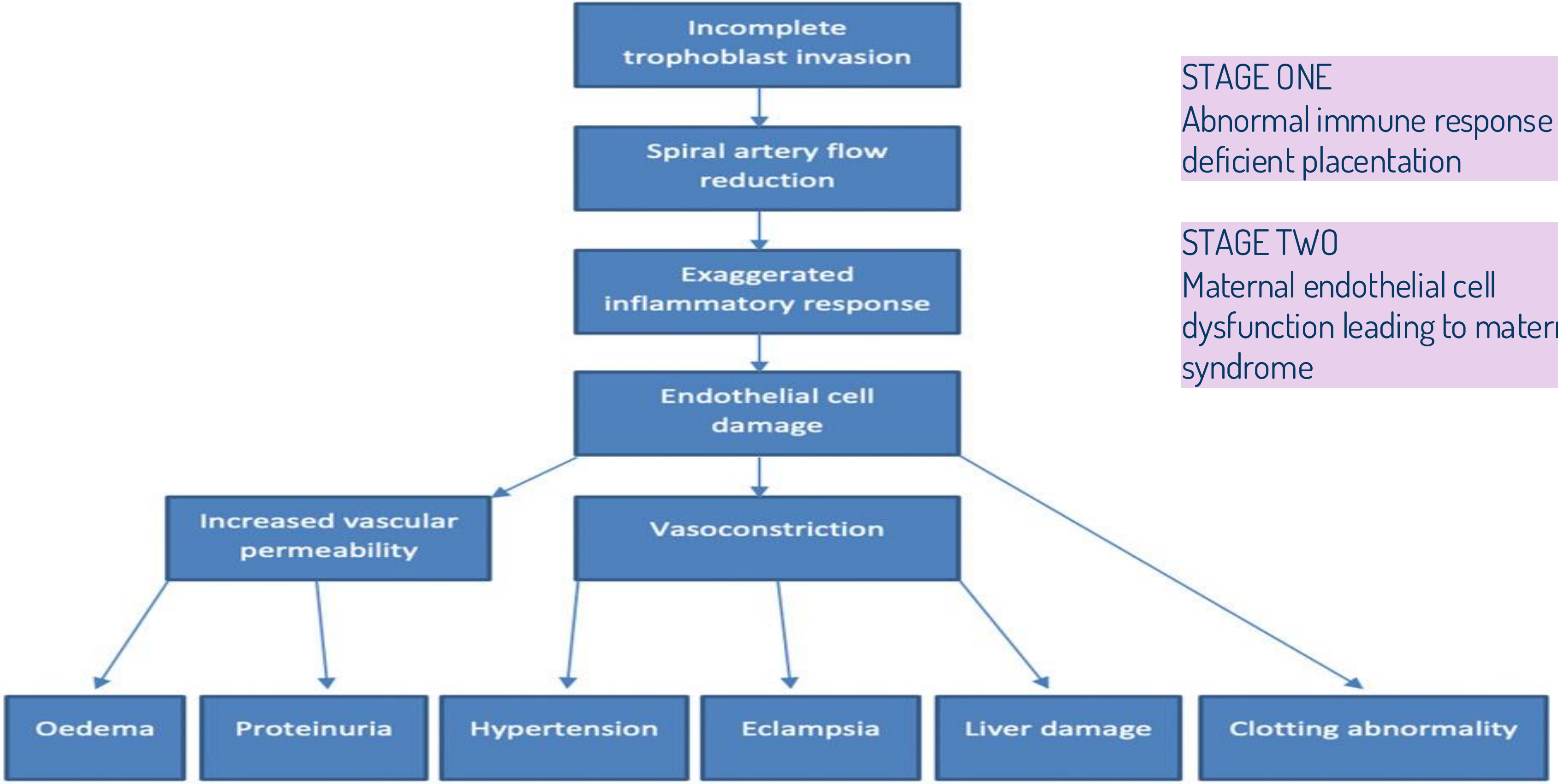
Pre-eclampsia

Pre-eclampsia is defined as new-onset gestational hypertension (systolic blood pressure ≥140 mmHg and/or diastolic blood pressure ≥90 mmHg) associated with new onset of at least one of the following:

- Proteinuria > 300 mg/24 hours
- Maternal organ dysfunction (liver, neurological, haematological or renal involvement)
- Uteroplacental dysfunction at or after 20 weeks' gestation


Normal
Syncytiotrophoblast
Deoxygenated blood
Maternal vein
Myometrium
Cytotrophoblast stem cell
Fetal blood vessel
Cytotrophoblast
Maternal blood
Blood flow
Tunica media smooth muscle
Anchoring villus
Spiral artery
Decidua
Floating villus
Anchoring villus cytotrophoblast column
Cytotrophoblast
Endovascular cytotrophoblast
Maternal endothelial cell
Preeclampsia
Deoxygenated blood
Myometrium
Maternal blood
Blood flow
Spiral artery
Decidua

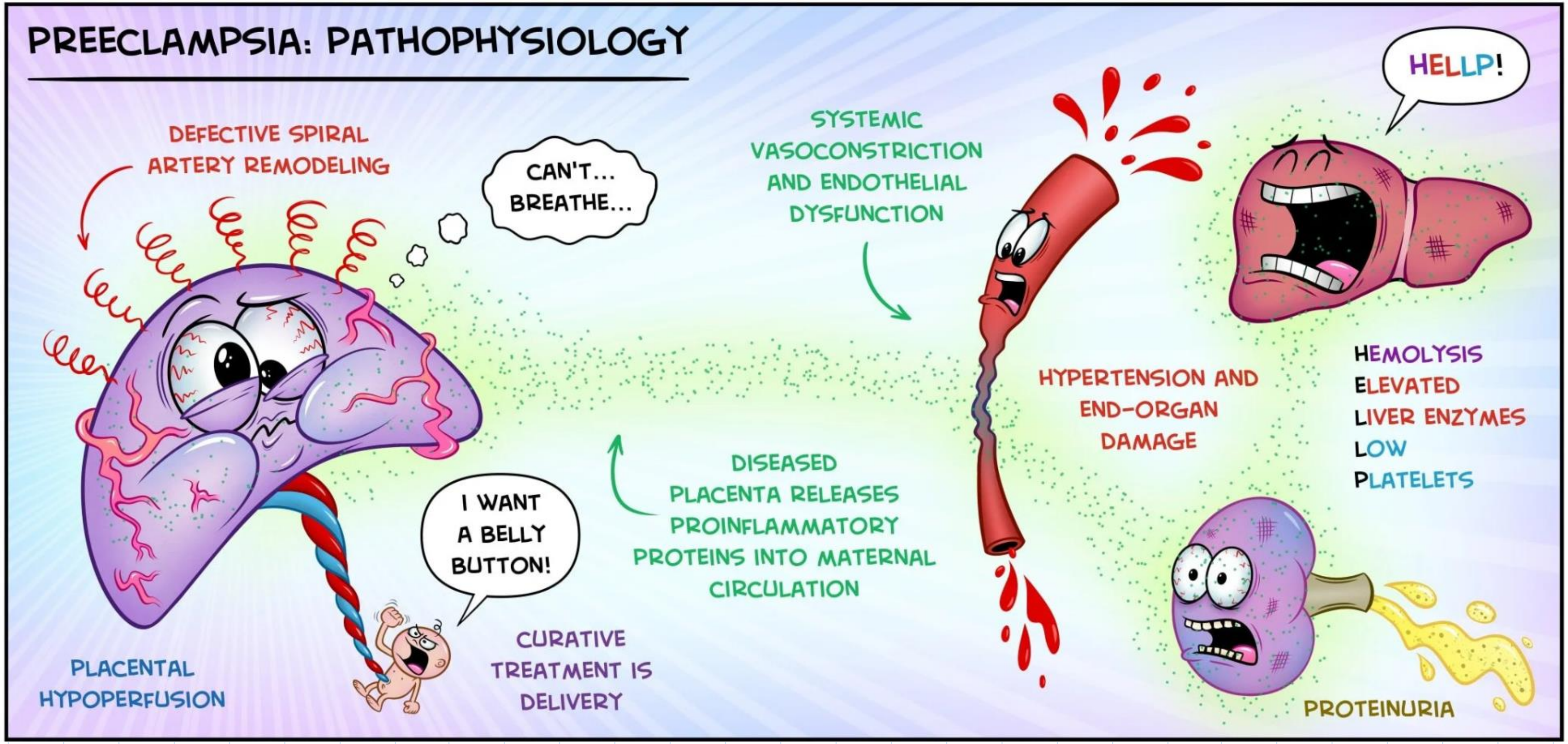
Pathophysiology



STAGE ONE
Abnormal immune response and deficient placentation

STAGE TWO
Maternal endothelial cell dysfunction leading to maternal syndrome

Preeclampsia: Pathophysiology



Risk factors for preeclampsia

Major risk factors

- ✓ Pre eclampsia in any previous pregnancy
- ✓ chronic kidney disease
- ✓ Autoimmune disease such as SLE or APS
- ✓ Chronic hypertension
- ✓ Type 1 or 2 Diabetes

Moderate risk factors

- ✓ First pregnancy
- ✓ Aged 40 years or older
- ✓ Pregnancy interval more than 10 years
- ✓ BMI of 35 or more at first visit
- ✓ Family hx of pre eclampsia
- ✓ Multi fetal pregnancy

Clinical Presentations

Visual disturbance

Frontal Headache

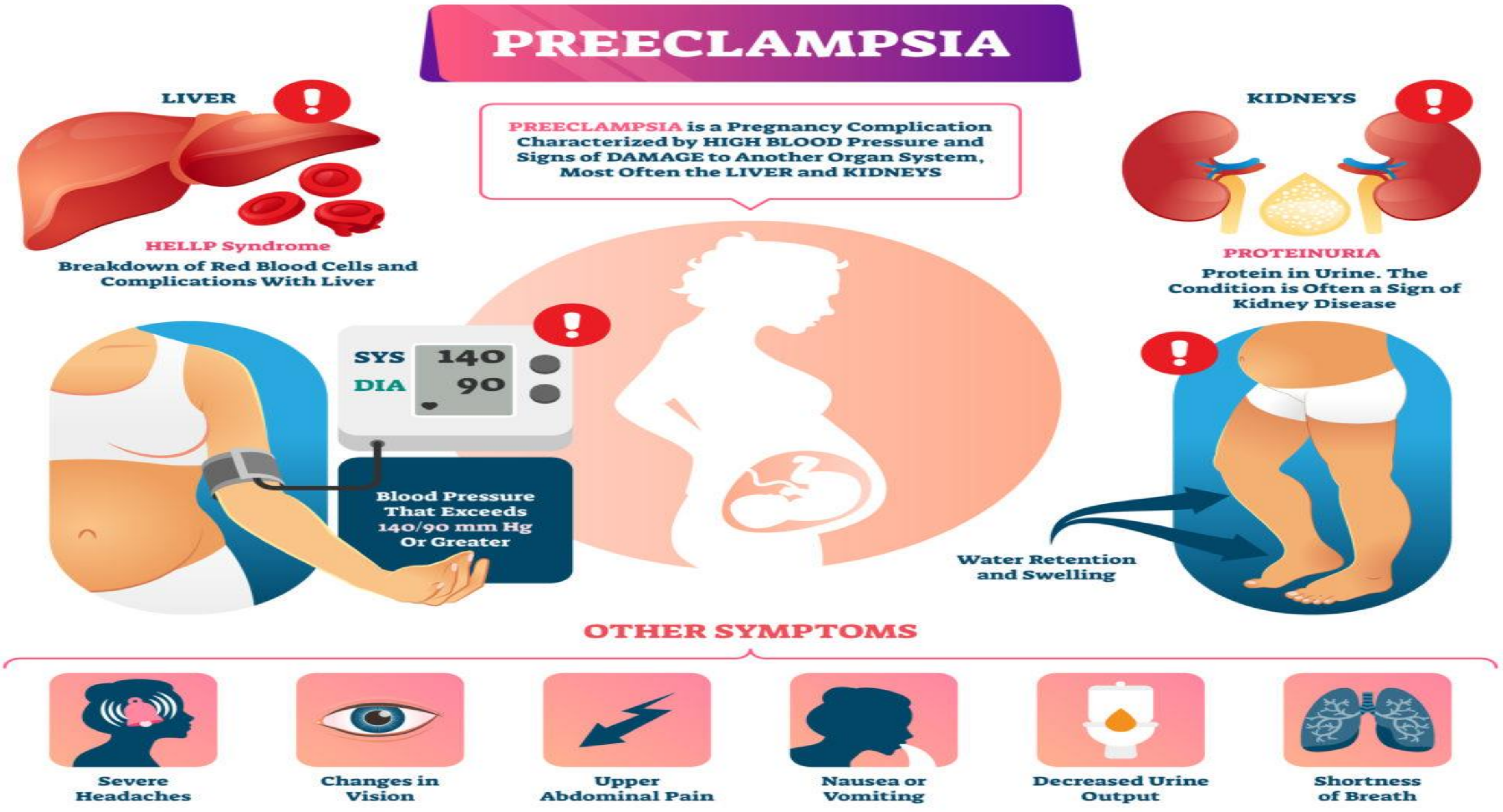
Epigastric pain

Clinical Presentations Continued

- The majority of pregnant women with pre-eclampsia are **Asymptomatic** or merely complain of general vague 'flu –like' symptoms.

In the clinical examination :

- Hypertension is usually the first sign,
- Edema of the face and hands may suggest pre-eclampsia
- Tenderness in epigastric region is a **Warning sign**
- Neurological examination may reveal hyperreflexia and clonus in severe cases.
- Urine testing for protein should be considered part of the clinical examination


PREECLAMPSIA
LIVER
HELLP Syndrome
Breakdown of Red Blood Cells and Complications With Liver
PREECLAMPSIA is a Pregnancy Complication Characterized by HIGH BLOOD Pressure and Signs of DAMAGE to Another Organ System, Most Often the LIVER and KIDNEYS
KIDNEYS
PROTEINURIA
Protein in Urine. The Condition is Often a Sign of Kidney Disease
SYS 140
DIA 90
Blood Pressure That Exceeds 140/90 mm Hg Or Greater
Water Retention and Swelling
OTHER SYMPTOMS
Severe Headaches
Changes in Vision
Upper Abdominal Pain
Nausea or Vomiting
Decreased Urine Output
Shortness of Breath

Investigations

Maternal monitoring:

- **CBC**: watch for falling plt and rising Hematocrit
- **Renal profile**: renal function and uric acid
- **Liver profile**
- Repetitive Testing for proteinuria

Fetal monitoring:

- **Ultrasound**: Asess fetal growth and amniotic fluid volume
- **Doppler** : evaluate uteroplacental and fetal blood flow
- **CTG** : Asess fetal well being ; loss of variability or deceleration > hypoxia

Complications

Maternal Complications	Fetal Complications
CNS involvement (eg. Seizure , stroke)	Intrauterine growth restriction (IUGR)
Disseminated intravascular coagulopathy (DIC)	Placental infarct and intrauterine fetal death (IUD)
Liver failure	Placental abruption
Renal failure	Prematurity
HELLP syndrome	Perinatal death
Antepartum and postpartum hemorrhage	Future cardiovascular diseases
High rate of C/S	
Future risk of chronic hypertension	
Death	

Managements

Management

There is no cure for pre-eclampsia other than to end the pregnancy by delivering the baby and placenta.

The principles of Mx of preeclampsia are;

- Early recognition of symotomless syndrome
- Awareness of the seriousness nature of the condition in its severest form
- Adherence to the guideline (admission, investigation, use of antihypertensive and anticonvulsant therapy
- Well timed delivery to prevent complications
- Postnatal follow up and counselling

Management depending on the degree of pre eclampsia

Outpatient Management of pre eclampsia

- Appropriate if:BP <160 systolic and <110 diastolic and can be controlled
- no or low (≤1+/<300mg/24h) proteinurea
- asymptomatic.
- Difficult to distinguish from gestational hypertension.

- Warn about development of symptoms
- BP and urine check :1-2 times per week
- blood biochemistry:weekly

Mild to moderate preeclampsia

BP <160 systolic and <110 diastolic with significant proteinuria and no maternal complications.

•Once significant proteinuria occurs, admission is advised:

≥ 2+ protein

>300mg proteinuria/24h

Blood pressure	Every 4 hour
Urine and dipstick	Daily, 24h collection for protein
Blood test	Every 2-3 days unless symptoms or signs worsen
Fetal monitoring	Daily CTG
Ultrasound	• Fortnightly (growth) • Twice weekly:Doppler /liquor depend on severity of pre-eclampsia
Antihypertensive therapy	If BP increas increases (>160 SYSTOLIC OR >110 DIASTOLIC)antihypertensive therapy shoud be used

Severe pre eclampsia

Defined as the occurrence of BP≥ 160 systolic or ≥110 diastolic in the presence of significant proteinuria (≥ 1g/24h or ≥2+ on dipstick), or if maternal complications occur.

Management

Blood pressure

- ⚠ BP needs to be stabilized with antihypertensive medication (must aim for <160 systolic *and* <110 diastolic).
- Initially use PO nifedipine 10mg: can be given twice 30min apart.
- If BP remains high after 2–3 nifedipine doses:
  - start IV labetalol infusion
  - ↑ infusion rate until BP is adequately controlled.
- Start maintenance therapy, usually labetalol; methyldopa if asthmatic.

Other management

- Take bloods for FBC, urea and electrolytes (U&E), LFTs, and clotting profile.
- Strict fluid balance chart: consider a catheter.
- CTG monitoring of fetus until condition stable.
- Ultrasound of fetus:
  - evidence of IUGR, estimate weight if severely preterm
  - assess condition using fetal and umbilical artery Doppler.

⚠ If <34wks, steroids should be given and the pregnancy may be managed expectantly unless the maternal or fetal condition worsens.

Treatments

Drug	Class /mechanism	Note	Side effect	Route	Emergency
Labetalol	Alpha and betta blocker	1st line drug according to nice guideline and safe in pregnancy		Oral /Iv	In severe cases of fulminating disease can be titrated rapidly
Nifidipine	Ca channel blocker	Rapid onset	Severe headache	Oral	
Methyldopa	Centrally acting agent	Long safety record-slow onset(>24hrs)	Sedation, depression	Oral	
Hydralazine	Arteiolar vasodilator	Used in severe fulminant cases – IV infusion titrated to BP		Iv	In severe cases of fulminating disease can be titrated rapidly

Hypertension in pregnancy: severe hypertension, severe pre-eclampsia and eclampsia in critical care

Medical management

- Measure BP hourly in women with hypertension, and every 15 to 30 minutes until BP is less than 160/110 mmHg in women with severe hypertension
- Treat women admitted to critical care during pregnancy or after birth immediately with one of:
  - labetalol (oral or intravenous)
  - oral nifedipine
  - intravenous hydralazine
- Continue appropriate ongoing antihypertensive treatment after initial management
- Monitor response to treatment to:
  - ensure BP falls
  - identify adverse effects for woman and fetus
  - modify treatment according to response
- If BP controlled within target ranges, do not routinely limit duration of second stage of labour
- If BP does not respond to initial treatment, consider operative or assisted birth

Fluid balance and volume expansion

In women with severe pre-clampsia:

- limit maintenance fluids to 80 ml/hour unless there are other ongoing fluid losses (for example, haemorrhage)
- do not preload with intravenous fluids before establishing low-dose epidural analgesia and combined spinal epidural analgesia
- do not use volume expansion unless hydralazine is antenatal antihypertensive; consider using 500 ml or less crystalloid fluid before or at same time as first dose of hydralazine in antenatal period.

Magnesium sulfate

- Give intravenous magnesium sulfate if a woman with severe hypertension or severe preeclampsia is having or has recently had an eclamptic fit.
- Consider giving intravenous magnesium sulfate if birth planned within 24 hours in woman with severe pre-eclampsia.
- Do not use diazepam, phenytoin or other anticonvulsants as alternatives to magnesium sulfate in women with eclampsia.

Regimen for magnesium sulfate

- Loading dose of 4 g given intravenously over 5 to 15 minutes, followed by infusion of 1 g/hour for 24 hours
- Further dose of 2 to 4 g given over 5 to 15 minutes if recurrent seizures

Nifedipine: at the time of publication (June 2019), some brands of nifedipine were specifically contraindicated during pregnancy by the manufacturer in its summary of product characteristics (SPC). Refer to the individual SPCs for each preparation of nifedipine for further details.

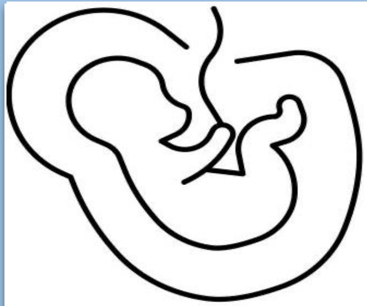
Magnesium sulfate regimen: also see The Eclampsia Trial Collaborative Group (1995) Which anticonvulsant for women with eclampsia? Evidence from the Collaborative Eclampsia Trial. Lancet 345:1455–1463.

The MHRA has issued a warning about the risk of skeletal adverse effects in the neonate following prolonged or repeated use of magnesium sulfate in pregnancy. Maternal administration of magnesium sulfate for longer than 5 to 7 days in pregnancy has been associated with skeletal adverse effects and hypocalcaemia and hypermagnesemia in neonates. If use of magnesium sulfate in pregnancy is prolonged or repeated, consider monitoring of neonates for abnormal calcium and magnesium levels and skeletal adverse effects.

NICE National Institute for Health and Care Excellence

This is a summary of advice on severe hypertension, severe pre-eclampsia and eclampsia in critical care from the NICE guideline on hypertension in pregnancy.

Additional management



Delivery

Depends on :

1. Gestational age
2. Severity of the condition
3. Prior obstetric history
4. Fetal condition

- Iatrogenic premature delivery is often required in severe preeclampsia
- Prior to 34 weeks of gestation, steroid should be given intramuscularly to the pregnant woman to reduce the chance of neonatal respiratory distress syndrome
- Delivery before term is often by cesarean section
- Such patients are at particularly high risk for thromboembolism and should be given prophylactic subcutaneous heparin and issued with anti thromboembolic stockings.

Postnatal

- the maternal blood pressure often decrease
- careful surveillance is required as it tends to increase again on the third or fourth postpartum day
- Breastfeeding is encouraged- contraception counselling
- medication should be changed to those drugs that are considered safe
- All patients should be revived 6 - 12 weeks post partum

Eclampsia

Is A serious complication of pre-eclampsia, **characterized by one or more generalized seizures or coma** in the absence of other neurological disorders, the seizures can occur before, during, or after delivery, but are most common in the **first postpartum week.**

Warning signs of eclampsia

Pain	Neurological	Blood pressure
• RUQ tenderness • Epigastric pain	• Headache • Agitation • Hyperreflexia &clonus • Papilledema • Blurred vision	• Uncontrolled blood pressure • Facial swelling • Poor urine output

Management of eclampsia

Immediate action

- Call for senior help & emergency alert team
- Perform drs ABC
- Auscultate & administer O2
- Fluid mx

Seizure control

- Magnesium sulfate 1st line
- Loading dose:4g IV
- Maintenance : 1g/hour IV

Blood pressure

- First-line: Labetalol or Hydralazine
- Labetalol: 20 mg IV → repeat 40–80 mg every 10 min (max 220 mg), maintenance 40 mg/hr
- Hydralazine: 5 mg IV → repeat up to 15 mg; if still high, double dose or IV drip

Alt: Nifidipine

Monitoring

- RR
- Urine output
- Patellar reflex
- Antidote: 10ml of 10% calcium gluconate slowly via IV

Screening & prevention

Screening:

There is no reliable early screening test for pre-eclampsia. No biomarker has shown adequate accuracy for clinical use. Uterine artery Doppler may detect a notch and high resistance in high-risk women, but it is not useful for low-risk population screening.

Prevention:

•Low-dose aspirin (75–150 mg daily) modestly reduces the risk of pre-eclampsia in high-risk pregnancies.

•Calcium supplementation may also be beneficial, especially in women with low dietary calcium intake

•Anti oxidant like vitamin C and E, zinc and other are controversial.

Hypertension in pregnancy: planning care for women at moderate and high risk of pre-eclampsia

Risk factors for pre-eclampsia

Moderate

- First pregnancy
- Age 40 or over
- Pregnancy interval of 10 years or more
- BMI of 35 kg/m² or more at first visit
- Family history of pre-eclampsia
- Multi-fetal pregnancy

High

- Hypertensive disease during previous pregnancy
- Chronic kidney disease
- Autoimmune disease such as systemic lupus erythematosus or antiphospholipid syndrome
- Type 1 or type 2 diabetes
- Chronic hypertension

↓

2 or more moderate risk factors, or

1 or more high risk factors?

↓

Give aspirin 75 to 150 mg/day from 12 weeks until birth

If any previous:

- severe eclampsia
- pre-eclampsia needing birth before 34 weeks
- pre-eclampsia with baby's birthweight below the 10th centile
- intrauterine death
- placental abruption

↓

Do ultrasound fetal growth, amniotic fluid volume assessment, umbilical artery doppler velocimetry:

- start at 28 to 30 weeks, or at least 2 weeks before previous gestational age of onset of hypertensive disorder if less than 28 weeks
- repeat 4 weeks later.

Use cardiotocography only if clinically indicated

i

Although this use of aspirin is common in UK clinical practice, at the time of publication (June 2019), aspirin did not have a UK marketing authorisation for this indication.

Community pharmacies cannot legally sell aspirin as a Pharmacy medicine for prevention of pre-eclampsia in pregnancy in England. Aspirin for this indication must be prescribed. The prescriber should see the summary of product characteristics for the manufacturer's advice on use in pregnancy. The prescriber should follow relevant professional guidance, taking full responsibility for the decision. Informed consent should be obtained and documented.

See the General Medical Council's Prescribing guidance: prescribing unlicensed medicines for further information.

NICE National Institute for Health and Care Excellence

This is a summary of advice on planning care for women at moderate and high risk of pre-eclampsia from the NICE guideline on hypertension in pregnancy.

Pre-eclampsia / Eclampsia
Characterised by high blood pressure and protein leak in the urine
Severe headaches
A
Changes in vision
Untreated, the brain can swell causing seizures, stroke, and death.
Fatigue
Reduced fetal growth
Tummy pain
Upper limb swelling
High blood pressure
The placenta's blood supply develops abnormally, reducing flow to the baby and releasing inflammatory factors. The body increases blood pressure to try and overcome this.
Bruising easily
Damaged kidneys leak protein causing the body to retain fluid. This results in small amounts of urine, and limb swelling.
artibiotics

References

- Obstetrics by Ten Teachers 21st Edition
- OXFORD HANDBOOK OFOBSTETRICS AND GYNECOLOGY
- NICE guidelines


THANKS FOR HELPING ME OUT